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On Campus • NIT JALANDHAR

CYBERKNIIGHTS

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PROBLEM STATEMENT

Modern networks span cloud, on-prem, and hybrid systems, making them **complex** and **constantly changing**. Security teams often rely on isolated, text-heavy tools that produce fragmented results without context. As a result, small misconfigurations are analyzed separately instead of as linked threats, **creating blind spots** where attackers can combine weaknesses into major breaches.

Impact: Delayed detection, poor prioritization, and higher breach risk due to fragmented analysis.

PROPOSED SOLUTION

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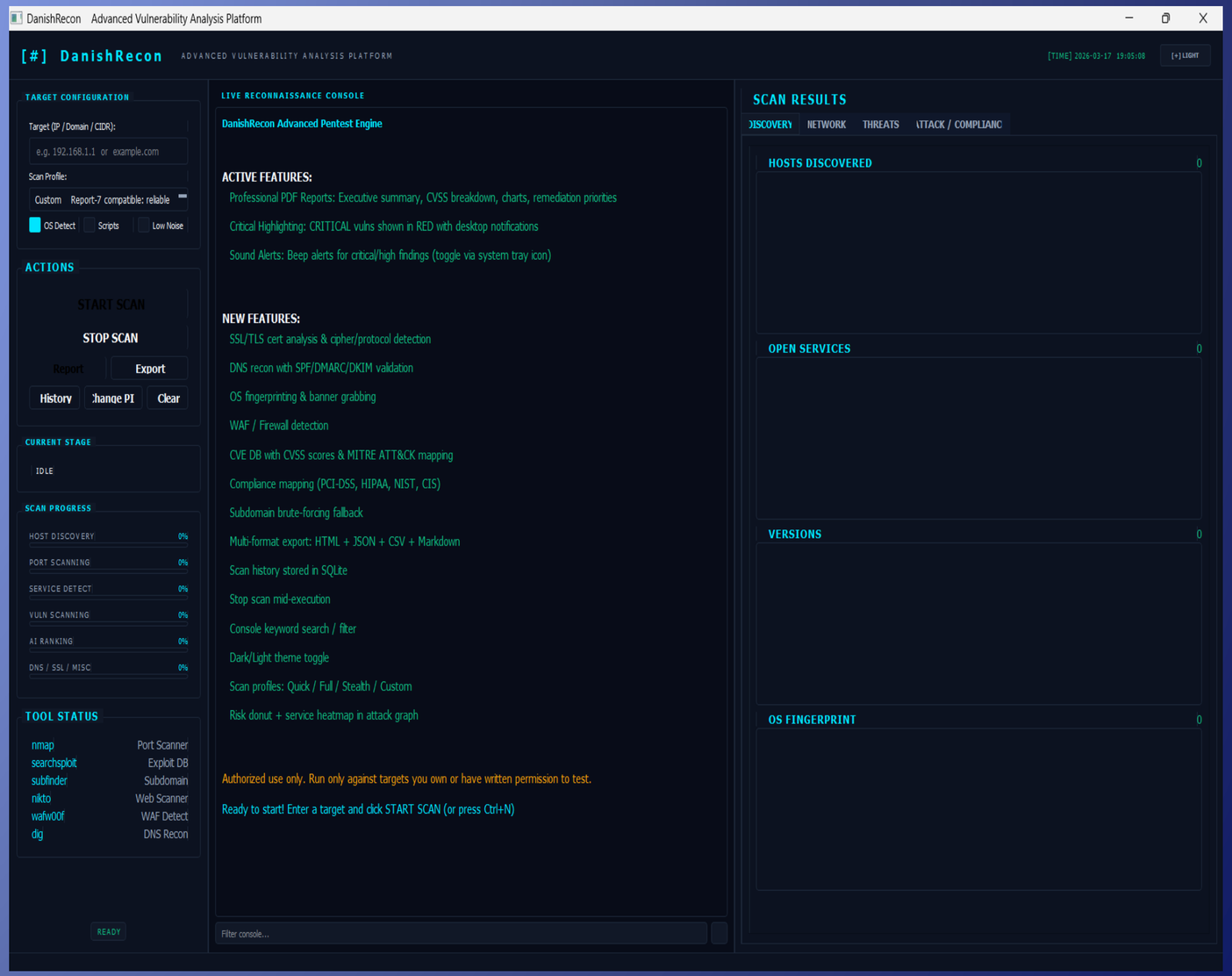
- **Unified Security Platform:** An all-in-one GUI application that automates fragmented security tools (Nmap, Nikto, DNS enumeration) into a unified, automated workflow..

- **Automated Threat Context:** Replaces raw data feeds with actionable intelligence by automatically mapping discovered services to known CVEs and assigning standard CVSS risk scores..

- **Visual Attack-Path Mapping:** Goes beyond text logs by generating interactive, visual attack-graph networks, allowing security teams to actually see how vulnerabilities chain together.

- **Instant Compliance Tracking:** Eliminates the manual bottleneck of compliance by automatically mapping discovered threats directly to PCI-DSS, HIPAA, NIST, and CIS frameworks.



FEATURES

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Dynamic Scanning & Reconnaissance

- **Targeted Profiles:** Quick, Full, Stealth, and Custom scan profiles tailored to specific engagement needs.
- **Deep Discovery:** Integrates **DNS recon** (SPF/DMARC/DKIM validation), OS fingerprinting, banner grabbing, and automated subdomain brute-forcing.

Vulnerability & Threat Intelligence

- **Context Engine:** Built-in CVE database with CVSS scoring and MITRE ATT&CK mapping.
- **Security Assessments:** Deep SSL/TLS protocol/**cipher analysis** and automated **WAF/Firewall** detection.

Analysis & Reporting

- **Compliance Ready:** Automatic compliance mapping against PCI-DSS, HIPAA, NIST, and CIS frameworks.
- **Professional Outputs:** Multi-format exports (PDF , HTML, JSON, CSV, MARKDOWN) featuring risk donuts, and attack vector distributions.

TECH STACK

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Front-End / User Interface

- **PyQt5:** Provides the robust, multi-threaded Graphical User Interface (**GUI)** and the seamless dark/light theme engine.

Core Application Logic

- **Python 3.11:** The backbone of the application, utilizing asynchronous subprocess and threading to manage heavy, concurrent scanning workloads without freezing the UI.

Data & Analytics Layer

- **NetworkX & Matplotlib:** Powers the active parsing of network paths and rendering of interactive, node-based Attack Network Graphs.
- **SQLite:** A lightweight, built-in relational database to persist scan histories, configurations, and state.

Security Tool Integrations (Sub-Processes)

- **Network & Web:** Nmap (port/service routing), Nikto (Perl-based web vulnerability scanning), and curl (banner grabbing).
- **Intelligence & OSINT:** Searchsploit (local exploit mapping), wafw00f (WAF detection), and subfinder (Go-based subdomain enumeration).

SQLite NMAP Nikto

WORKFLOW DIAGRAM

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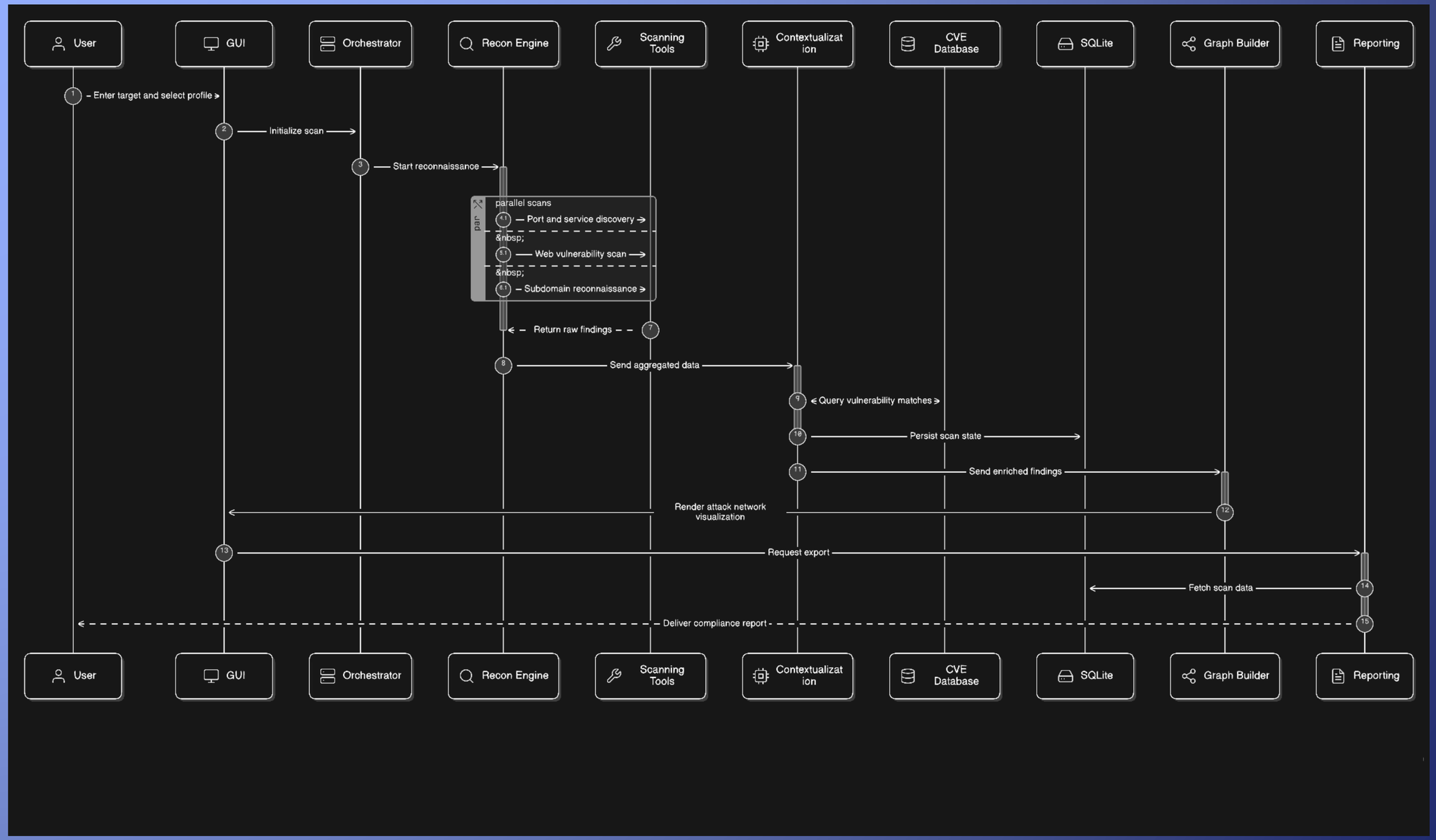

TARGET AUDIENCE AND ENGAGEMENT

1. Penetration Testers & Red Teams (Offensive)

- **The Value:** Speeds up the initial reconnaissance phase. Uses automated OSINT, port scanning, and exploit mapping to highlight immediate attack vectors without requiring manual terminal jockeying.

2. Security/SOC Analysts (Defensive)

- **The Value:** Provides a unified "single pane of glass" to monitor infrastructure. The visual Attack Graph allows analysts to see exactly how misconfigurations impact critical inner nodes.

3. Auditors & Compliance Officers (Administrative)

- **The Value:** Translates raw technical data (like outdated cipher suites) directly into recognizable compliance violations (PCI-DSS, NIST), automating the creation of executive PDF summaries.

4. Small/Medium Enterprises (SMEs)

- **The Value:** Organizations without the budget for $10,000+ enterprise scanners can use DanishRecon as an accessible, deeply capable alternative for identifying perimeter blind spots.

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FUTURE SCOPE

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Future Scope: Extending Protection & Reach

Our long-term vision involves expanding **DANISHRECON** capabilities through integration and enhanced protocols.

Real-Time Threat Intelligence Integration

Live CVE and exploit enrichment using **NVD + CISA KEV** APIs for **dynamic risk prioritization.**

Cloud and Enterprise-Scale Coverage

Multi-cloud asset discovery across **AWS, Azure, and GCP** with scheduled **continuous scanning.**

Advanced Intelligence and Automation

ML-assisted risk prediction to prioritize vulnerabilities based on **exploit likelihood** and impact